NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 10

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following are examples of hardware-based attacks?

a. Physical probing
b. SQL injection
c. Cross-site scripting
d. Electromagnetic emission analysis
e. Phishing

Correct Answer: a, d

Detailed Solution: Physical probing and electromagnetic emission analysis both require access to or measurement of the physical hardware running a computation. SQL injection, cross-site scripting, and phishing are software/application-level or social engineering attacks unrelated to hardware access.
Thus, the correct options are (a) and (d).

QUESTION 2:

Which of the following statement(s) is/are **true** regarding side-channel attacks?

a. They exploit weaknesses in the implementation of an algorithm, not the algorithm itself.
b. They may require physical access to (or close proximity to) the target device, depending on the side channel being exploited.
c. They rely solely on analyzing input/output pairs of the algorithm.
d. They can exploit information such as timing, power consumption, or electromagnetic emissions.
e. They exploit mathematical weaknesses in the cryptographic algorithm's design.

Correct Answer: a, b, d

Detailed Solution: Side-channel attacks exploit weaknesses in how an algorithm is implemented (hardware or software), rather than flaws in the underlying algorithm's mathematical design. Depending on the side channel being exploited, they may use information such as timing, power consumption, or electromagnetic emissions and may require physical access or close proximity to the target device. They do not rely solely on input/output analysis, which is characteristic of black-box analysis.
Thus, the correct options are (a), (b), and (d).

QUESTION 3:

For modular exponentiation of x^{22} using the square-and-multiply method, how many squaring and multiplication operations are required, respectively?

a. 4 and 2
b. 3 and 2
c. 4 and 3
d. 5 and 2
e. 3 and 3

Correct Answer: a

Detailed Solution: The binary representation of 22 is 10110. Number of squaring's = total bits - 1 = 5 - 1 = 4. Number of multiplications = (number of 1's) - 1 = 3 - 1 = 2. Hence, 4 squaring's and 2 multiplications are required.
Thus, the correct option is (a).

QUESTION 4:

Which of the following statement(s) is/are **true** regarding Differential Power Analysis (DPA) and Simple Power Analysis (SPA)?

a. DPA requires only a single power trace measurement.
b. DPA requires multiple power trace measurements combined with statistical analysis.
c. DPA uses statistical analysis of multiple traces to reveal information that may not be apparent from individual traces.
d. DPA is generally less effective than SPA.
e. SPA relies on directly observing patterns in a single or few traces.

Correct Answer: b, c, e

Detailed Solution: DPA uses multiple power measurements combined with statistical analysis to identify correlations between power consumption and sensitive data. In contrast, SPA relies on directly observing distinguishable patterns in a single or small number of traces. DPA is therefore useful for revealing leakage that may not be apparent from individual traces.
Thus, the correct options are (b), (c), and (e).

QUESTION 5:

Which property of a PUF refers to the requirement that, given a PUF and an input (challenge) x, its corresponding output (response) y = PUF(x) can be computed efficiently?

a. Unclonability
b. One-wayness
c. Evaluability
d. Unpredictability
e. Tamper-evidence

Correct Answer: c

Detailed Solution: Evaluability refers to the property that computing the response y = PUF(x) for a given challenge x should be computationally easy/efficient. Unclonability refers to the difficulty of constructing a different device that mimics the same challenge-response behaviour, and one-wayness refers to the difficulty of finding x given only y.
Thus, the correct option is (c).

QUESTION 6:

_____ is the full form of the acronym PUF used in hardware security.

a. Physically Unique Function
b. Physically Unclonable Function
c. Protected Unclonable Feature
d. Physically Untraceable Function

Correct Answer: b

Detailed Solution: PUF stands for **Physically Unclonable Function**, a hardware security primitive that exploits inherent manufacturing variations in a device to generate unique, device-specific responses.

Thus, the correct option is (b).

QUESTION 7:

What is a hardware Trojan?

a. A malicious software virus that specifically targets firmware update files.
b. A malicious, intentional modification to a chip's circuitry that activates under specific trigger conditions.
c. A testing tool used to validate hardware functionality before shipping.
d. A feature designed to make hardware run faster under heavy load.

Correct Answer: b

Detailed Solution: A hardware Trojan is a malicious modification embedded within a chip's circuitry, remaining dormant until a specific triggering condition is met, at which point it activates an unintended, harmful operation (the "payload").
Thus, the correct option is (b).

QUESTION 8:

Which of the following statement(s) is/are **true** regarding hardware Trojans?

a. They typically incur small hardware overhead, making them difficult to detect.
b. They are generally easy to detect using a single, universal detection method.
c. Detecting hardware Trojans often introduces additional design, testing, or runtime overhead.
d. Hardware Trojans are exclusively inserted through software-level modifications.

Correct Answer: a, c

Detailed Solution: Hardware Trojans typically add minimal overhead to remain stealthy, and no single detection method can reliably catch all Trojan variants, with detection efforts often adding their own overhead. No universal, easy detection method exists, and Trojans are inserted at the hardware (circuit) level, not through software modifications.
Thus, the correct options are (a) and (c).

QUESTION 9:

Which of the following hardware attack(s) are generally considered invasive in nature?

a. Black-box testing
b. Physical probing
c. Side-channel power analysis
d. Timing analysis

Correct Answer: b

Detailed Solution: Physical probing and reverse engineering that involves decapping or delayering a chip requires direct physical alteration of the device, making them invasive. Black-box testing, side-channel power analysis, and timing analysis are typically non-invasive, relying on external measurements or input/output observations rather than physically opening the chip.
Thus, the correct option is (b).

QUESTION 10:

Which of the following software-level technique(s) can help mitigate timing-based side-channel attacks?

a. Introducing redundant/dummy computations to equalize execution time across branches.
b. Masking sensitive intermediate data values during computation.
c. Deliberately increasing overall program complexity without addressing timing variance.
d. Ensuring constant-time implementations for security-critical operations.

Correct Answer: a, d

Detailed Solution: Introducing redundant or dummy computations to equalize execution time across different execution paths and using constant-time implementations for security-critical operations can help mitigate timing-based side-channel attacks. Simply increasing program complexity without addressing timing variations does not prevent timing leakage. Masking sensitive intermediate values is primarily used to mitigate other forms of side-channel leakage and does not, by itself, eliminate timing leakage.
Thus, the correct options are (a) and (d).

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